

Hamza Waqas

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Education

City St George's, University of London
BEng Aerospace Engineering (*Predicted First Class with Honours*)

London, UK
Graduation July 2027

Experience

Panasonic Avionics Corporation

Summer Work Experience

July 2025

London, UK

- Looked into integration of IFE systems onto Airbus aircraft and the current issues they had on the customer aircraft.
- Followed tests taken to recreate the power interrupts from switching from APU to GPU or to engine power.
- Studied older IFE systems facing issues to draw parallels and determine if these were hardware or aircraft related.
- Shadowed the engineers working on the project, covering part of the engineering investigations and execution of the aircraft test plan.

Formula Student UK

Head of Mechanical Systems

September 2024 – Present
Extracurricular in 1st and 2nd year

- Led chassis design and structural validation for the team's first EV and car in 8 years, targeting the 2027 season.
 - Designed the full front and rear suspension geometry from a clean-sheet chassis, defining roll centre height, camber, anti-dive, and pushrod/bellcrank motion ratio targets.
 - Validated structural integrity to 40g frontal/side impact loading while cutting chassis mass from 57.8kg to 53.2kg (safety factor >1.6); delivered full vehicle design-to-build in 10 months against fixed facility constraints.
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Projects

CAGE (Confined Aerial Geometry) - AEW&C system for UAVs

- Designed the complete software and firmware architecture for a radar sensing system built from four independent units across dual ESP32-S3 controllers, with C++ firmware handling sensor acquisition and live UDP telemetry over Wi-Fi.
- Built a Python fusion and visualization pipeline that ingests live UDP streams from multiple sensors, merges detections, and renders a unified 360° polar display with overlapping coverage in real time.
- Implemented and tested the full pipeline from sensor to display on a standalone rig, validating telemetry reliability, detection coverage, and tracking of multiple targets ahead of integration.

ML - Based Gas Turbine Emissions Modelling

- Built a complete Python machine learning pipeline (scikit-learn) predicting gas turbine CO and NOx emissions from a 35,499-row sensor dataset, including data cleaning (missing values, duplicates, IQR based outlier removal) and feature engineering (humidity-temperature ratio, temperature-pressure interaction) informed by correlation analysis.
- Trained and compared four models, polynomial, ridge, and lasso regression plus a tuned gradient boosting regressor, evaluated with MAE, RMSE, and R^2 on a fixed train, validation, and test split, with gradient boosting performing best.
- Achieved a held-out test R^2 of 0.71 for CO and 0.65 for NOx, built the matplotlib visualizations (heatmaps, boxplots, predicted versus actual plots) used to interpret features and validate performance, and authored the full technical report.

Spacecraft ADCS Simulator (Attitude Determination and Control System)

- Built a full 6-DOF simulation of a spacecraft's attitude dynamics and control system in MATLAB from first principles, modelling sensor, actuator, and disturbance behaviour as interchangeable modules, validated to approximately 10^{-13} relative accuracy against closed-form solutions.
 - Implemented three interchangeable attitude controllers (PID, LQR, LQI), building the LQR/LQI Riccati solver from scratch and verifying it to 2×10^{-10} agreement against MATLAB's native solver and independently derived solutions.
 - Designed a Monte Carlo validated Kalman filter estimator and a supervisory state machine (SAFE, ACQUIRE, NOMINAL) for autonomous fault recovery, quantifying a 4.2x pointing error increase under realistic uncertainty and recovering performance within approximately 260 seconds without mode chattering.
 - Currently porting over to python (open sourced) with an extended suite of features to make it more attainable and education focused for those interested in spacecraft technology.
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Skills

Languages: Python (scikit-learn, Pandas, NumPy, Matplotlib), C/C++ (embedded systems, Arduino/ESP32), MATLAB

Tools and Technologies: Git, VS Code, Ardupilot, Mavlink, SolidWorks/FEA, ANSYS, ABAQUS, SIEMENS NX/FEA, Fusion 360, XFLR5, 3D Printing, Microsoft Office, Google Suite